

Push swap

LIS method

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What is LIS ?

LIS = longest increase sequence

LIS

3	1	5	6	2	4	7
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In this example the LIS is 3567

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So I have to push the element who's not in LIS in stack B

A
3
5
6
7

I have to find the sempai and calculate the cost
for send it to stack A in the right position

Sempai = the value bigger and most close of the value in stack B

Cost = distance with the top of stack B + distance of the
sempai with the top

Let me show you a
example

B
1
2
4

A
3
5
6
7

The sempai of 4 is 5 because the value
who are bigger and close of 4 in stack_a it
5

For calculate the cost of 4 I will do :
The number of move for bring 4 on the top is 1 , because I
need to do a RRA. The number of move for bring 5 on the
top is 1 because I just need to do a RA.
Cost = 1 + 1 = 2

Becarful because there is a other case

B
1
2
4

Sempai = 3
Cost = 0

Sempai = 3
Cost = 1

Sempai = 5
Cost = 2

A
3
5
6
7

This case is when I have a value who don't have a sempai.

For example : the value five don't have a sempai in stack A, in the case I will attribute him the smaller value in stack A for sempai.

Cost of 8 = 1, because i need to bring 8 on the top and 3 is already on the top.

Cost of 8 = 1 + 0 = 1

It a example

B
1
2
4
8

Sempai = 3
Cost = 0

Sempai = 3
Cost = 1

Sempai = 5
Cost = 2

Sempai = 3
Cost = 1

A
3
5
6
7

Once I determinate the sempai and the cost of all value
in stack B

I will select the value with the smallest cost and push it
on stack_a

Here I will send 1

B
1
2
4

Sempai = 3
Cost = 0

Sempai = 3
Cost = 1

Sempai = 5
Cost = 2

A
1
3
5
6
7

When I push 1 value, I calculate again the cost
and I search the sempai

And I do this since my stack B isn't empty

B
2
4

Sempai = 3
Cost = 1

Sempai = 5
Cost = 3

A
4
5
6
7
1
2
3

My stack B is empty now, and my stack A it look like this

And is not really sort because I have a 4 on the top but I have to bring the smallest value on the top

So I use a function for see if it better to do a RRA or a RA for bring the smallest value on the top

Now I do a RRA or a RA until the smallest value go on the top

A
1
2
3
4
5
6
7